

Question Decomposition Prompt

Prompt:
You are an intelligent assistant tasked with decomposing nested questions into a plan of several sub-questions using the following question taxonomy.

We have four types of unit questions, where s is the subject, p is the predicate, o is the object, and * means the missing part. Below, we define the four types of questions with some concrete examples for each type.

The four types of questions are:

- **<s,*,*> (0):**
In this type, the object and concrete predicate are both missing in the facts and question asks about the general or broad information of the subject (e.g., themes, concepts, or a general description).
Examples:
 - "Tell me about 'The Woman in Black: A Ghost Play'."
 - "Give a broad description about 'Frankenstein, or The Modern Prometheus'."
 - "Who is 'Barack Obama'?"
- **<s,p,*> (1):**
In this type, the object is missing in the facts and question focuses on finding the object, with specific and concrete relations and attributes to the subject are provided by a concrete predicate p.
Note that this type of question can have multiple predicates and subjects, and containing time or counting constraints in the predicates.
Examples:
 - "Who are the authors of the book 'Sunshine for the Latter-Day Sa'?"
 - "What series have the author of the book 'Cookies for the Dragon (Saint Lakes, #2.1)' published?"
 - "What are the 5 biggest cities in the usa?"
- **<s,*,o> (2):**
In this type, both the subject and object are given but the predicate is missing. Question asks about their relationships or interactions (e.g., conceptual connections, influences).
Examples:
 - "What is the relationship between 'Kizuna' and 'Kazuma Kodaka'?"
 - "What are the impacts of information communication technology to public relation practices?"
 - "How are the directions of the velocity and force vectors related in a circular motion?"
- **<s,p,o> (3):**
In this question, the subject, predicates object are all given. Questions either inquires about particular aspects of the relationship between the entities or verifies whether a specific relationship exists.
Examples:
 - "Have the authors 'Rubem Fonseca' and 'Lygia Fagundes Telles' ever published books in the same publishers?"
 - "Do the publishers 'Scholastic Inc.' and 'Klutz' have any authors publishing books in both of them?"

****Instruction:****
When given a nested question that can not be directly classified into one of the four types, decompose it into a plan of several sub-questions with each being a unit question based on the definitions above.

We will also provide you the graph schema, which indicates the types of nodes and edges, and by what relations nodes are connected. This will help you to identify the sub-questions and generate a description for each step of the decomposition plan.

For each step of the decomposition plan, you should first identify the type of the sub-question and then generate a description which will further guide the instantiation of the concrete sub-question.

For example:
Question: "Tell me about the academic contributions of the academic collaborators of the scholar 'L. Foldy'."
Decomposition:
1. <s,p,*>: find the academic collaborators of the scholar 'L. Foldy'.
2. <s,*,*>: retrieve the academic contributions of the scholars found in the previous step.

Important Note: Always avoid consecutive <s,p,*> steps, they can be merged into one single <s,p,*> question with a chain of relations (multi-hop), and you should merge them into one step, just like the examples shown in demonstration of the <s,p,*> question ("Who are the authors of the books that are published by the publishers that have published books of the series 'Shifter Justice'?").

****Graph Schema:****
{graph_schema}

****Question:**** {query}

Return your answer in the following format:
``plan
1. question-type: description 1
2. question-type: description 2
:::